

T3 matched EFT Lagrangian terms: EFT_2_after_S1_S2

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out S1, S2. Remaining active T3 fields: none.

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	H	ℓ	e	F	H	ℓ
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0								
cont.															
cont.															
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0								
cont.															
cont.															
T3-A	0	1	0	3	1	2	$\frac{1}{2}$								
cont.															
cont.															

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$H \ell e$	$F H \ell$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{\sqrt{3}\epsilon M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$	$\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{\sqrt{3}M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$
								$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{\sqrt{3}M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$	$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{\sqrt{3}M_F}{\sqrt{3}M_F}$
								$+\frac{2h\lambda_{T3}L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}}{\sqrt{3}\epsilon M_F}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$	$+\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}}}{\sqrt{3}M_F}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$
cont.								$+\frac{2h\lambda_{T3}L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}}{\sqrt{3}M_F}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}{4\epsilon} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
cont.									$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}{4\epsilon} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}{4\epsilon} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{3}{8}h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}{4\epsilon} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{\epsilon M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$	$\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$
								$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$	$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{M_F}{M_F}$
								$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}}{\epsilon M_F}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$	$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}}}{M_F}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$
cont.								$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}}{M_F}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}{4\epsilon} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
cont.									$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}{4\epsilon} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}{4\epsilon} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}}{4\epsilon} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	

Field combinations 2 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell e$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$